Privacy-Preserving Human Activity Recognition from mmWave Radar Point Clouds using Lightweight Point Transformers

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***Abstract*—Millimetre-wave (mmWave) radar can recognise human activities without capturing images of people. Its point clouds, however, are extremely sparse and full of clutter, and the accuracies reported for them depend heavily on how the data are split. We present *STPT-Lite*, a lightweight spatio-temporal point transformer that applies self-attention among the points of short temporal segments and then across segment tokens. Each token is enriched with an explicit radar motion descriptor, and the model needs neither farthest-point sampling nor k-NN search. Doppler-based person centring and spatial gating remove the static clutter that makes up about half of the radar returns. With knowledge distillation from a larger teacher, noise-robust training and INT8 weights, STPT-Lite has 117.5 K parameters, needs 36.6 MFLOPs per 2-s window, occupies 0.14 MB and runs in 2.58 ms on one CPU thread. We evaluate it against six re-implemented baselines (voxel CNN-LSTM, PointNet, PointNet+BiLSTM, PointNet++, DGCNN and Point Transformer) on two public datasets with three seeds and leakage-free splits. STPT-Lite reaches $96.83 \pm 0.69\%$ on the official RadHAR test recordings and $49.56 \pm 1.98\%$ top-1 (78.03% top-3) on 49 MiliPoint actions performed by subjects never seen in training, against 32.57% for the best baseline. Under a random window split, which is leaky, the same architecture scores 79.48%, showing how strongly the evaluation protocol inflates reported accuracy. Ablations, cross-dataset transfer, robustness tests and attention/occlusion analyses explain where the accuracy comes from.**

***Index Terms*—mmWave radar, point clouds, human activity recognition, privacy-preserving sensing, point transformer, knowledge distillation, edge AI**

I. Introduction

Monitoring daily activity at home, for example in elderly care, fall-risk screening or rehabilitation, needs sensing that works in the dark, through clothing, and without recording images of people. Cameras fail the last requirement, and wearables depend on users remembering to wear and charge them. Low-cost single-chip mmWave radars (60–81 GHz) output a sparse 3-D point cloud with a Doppler velocity for every point. Identity is not visually recoverable from such data, which makes radar attractive for privacy-preserving human activity recognition (HAR) [1], [2].

Radar point clouds differ sharply from the LiDAR and CAD data that most point-cloud networks are designed for. A frame contains only about 6–21 returns (Section IV-A). Many returns are static clutter or multipath rather than the person. The useful signal lies in how the cloud moves over time rather than in its instantaneous shape. Existing approaches either voxelise the cloud [1], which wastes computation on empty cells, or apply dense-point backbones such as PointNet++, DGCNN or Point Transformer [3]–[5]. Those backbones rely on farthest-point sampling and k-NN search, which are slow on embedded processors. A second, less visible problem concerns evaluation. Radar HAR samples are short windows cut with large overlap from long recordings, so random window-level splits place near-identical samples in training and test. Published accuracies on the same dataset consequently range from about 34% to 98% [6], [7].

This paper studies privacy-preserving HAR from mmWave point clouds with a focus on two things: models that are light enough for edge deployment, and measurements that are honest. Our contributions are:

- **STPT-Lite**, a factorised spatio-temporal point transformer. Attention runs within short temporal segments and then across segment tokens, each token is augmented with an explicit radar motion descriptor, and no FPS or k-NN is needed. With knowledge distillation, noise-robust training and INT8 weights it reaches 96.83% on RadHAR with 117.5 K parameters, 36.6 MFLOPs, 0.14 MB of storage and 2.58 ms single-thread CPU latency.
- **Radar-specific pre-processing**: moving-point (Doppler-based) person centring and spatial gating. Together they remove the static clutter that makes up about half of the returns, along with the scene shortcut that clutter creates.
- **Leakage-free evaluation** on two public datasets: recording-disjoint validation on RadHAR and a subject-disjoint split on MiliPoint, where STPT-Lite reaches 49.56% top-1 on 49 actions performed by unseen people. The same architecture scores 79.48% under a leaky random split, which quantifies how much the protocol inflates reported accuracy.
- A study with **six re-implemented baselines, three seeds, paired significance tests, ablations, MiliPoint→RadHAR transfer, robustness to seven controlled perturbation types, and explainability** (temporal attention, occlusion and point saliency).

Code, pre-processing and all run logs are released for reproducibility.

II. Related Work

mmWave point-cloud HAR. RadHAR [1] released the MMActivity data used here. It voxelised each frame into a $10\times 32\times 32$ grid and classified sequences with a time-distributed CNN and a BiLSTM (90.47% accuracy). Later work moved to models that consume points directly. MMPoint-GNN [8] builds dynamic graphs per frame. Lee and Kim [9] pre-train a GNN on 3-D joint coordinates and report 96.55% on RadHAR. Gu et al. [10] pair a light PointNet with a bidirectional linear-attention LSTM (95.12%, 79.7 K parameters). MAK-GCN [7] learns multi-head adaptive graph kernels (97.54%, 1.86 M parameters). OG-PCL [11] uses occupancy-gated convolutions (91.75%, 0.83 M parameters) and reports PCT+LSTM [12] at 92.78%. Gao et al. [13] model each frame as a star graph around a virtual centre. Lim et al. [14] deploy a 4-bit pillar-based classifier on an FPGA. For gestures, Pantomime [15] and Tesla-Rapture [16] showed that sparse radar point clouds support lightweight recognisers. The same sensing also drives rehabilitation [17], mesh reconstruction [18], [19] and scene flow [20]. Ullmann et al. [2] survey radar-based continuous HAR.

Benchmarks and evaluation protocols. MiliPoint [6] (49 actions, 11 subjects) and MM-Fi [21] are recent large-scale benchmarks. On MiliPoint action classification the original authors report at most 34.45% (PointNet++), whereas MAK-GCN reports 98.25% under an 80/10/10 split. Several RadHAR studies validate on 20% of training windows drawn at random, and those windows overlap by 50 of 60 frames. Data leakage of this kind is a known way to inflate results [22]. To our knowledge, no prior work has measured its effect on radar point-cloud HAR.

Point-cloud backbones. PointNet [23] and PointNet++ [3] introduced learning on raw point sets. DGCNN [4] builds dynamic k-NN graphs. Point Transformer [5], PCT [12], PTv2/PTv3 [24], [25], PointNeXt [26] and PointMLP [27] raised accuracy on dense LiDAR and CAD data. P4Transformer [28] handles point-cloud videos. All of these assume hundreds to thousands of points per frame and rely on FPS and k-NN, which are costly on micro-controllers.

Research gap. (i) Most radar HAR models either reuse dense-point backbones or voxelise the data. Few exploit what is specific to radar: extreme sparsity, a direct Doppler measurement, and static clutter. (ii) Reported accuracies are rarely obtained with leakage-free (recording- or subject-disjoint) splits, and protocols differ between papers. (iii) Cross-dataset transfer between radar datasets and edge cost (latency, size, quantisation) are seldom reported together. This work addresses all three gaps.

III. Methodology

A. Problem formulation

A frequency-modulated continuous-wave (FMCW) mmWave radar outputs, at every frame t, a small unordered set of detections $\mathcal{F}_t = \{\mathbf{p}_{t,i}\}_{i=1}^{n_t}$ with $\mathbf{p} = (x, y, z, v, \iota)$: the 3-D position, the radial (Doppler) velocity v and the reflected intensity ι. The count n_t is small and varies (median 21 on RadHAR and 6 on MiliPoint, Section IV-A). Given a window $\mathcal{W} = (\mathcal{F}_{t_0}, \ldots, \mathcal{F}_{t_0+T-1})$ of T frames (2 s), we learn $f_\theta : \mathcal{W} \mapsto \hat{y} \in \{1, \ldots, K\}$ that predicts the activity. f_θ must be invariant to the order of points within a frame, robust to missing detections, and cheap enough to run on an edge device at the radar frame rate. The radar never records an image of the person, so the sensing is privacy-preserving by construction.

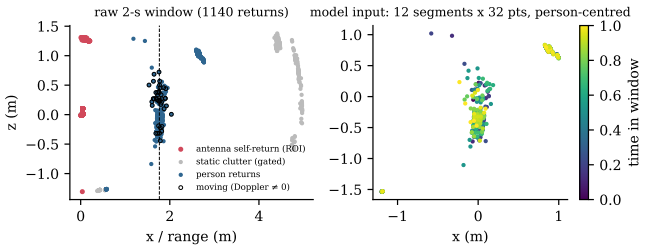


Fig. 1. A raw 2-s RadHAR window (left): antenna self-returns (red), static clutter removed by gating (grey), person returns (blue) and moving returns (circled). Right: the resulting person-centred model input, coloured by time.

B. Radar-specific pre-processing

Raw RadHAR windows contain three kinds of returns (Fig. 1): antenna self-returns at $x \approx 0$, static multipath and clutter several metres behind the subject, and the person. We apply three steps. **(i) ROI**: drop returns with $x < 0.3$ m (34.6% of all points). **(ii) Person centring**: the window centre $\mathbf{c}$ is the median position of the *moving* returns ($v \neq 0$), because static clutter has zero radial velocity; if a window has no moving return we fall back to the median of all returns. Coordinates are expressed relative to $\mathbf{c}$, which removes the subject's absolute position and distance from the radar. **(iii) Spatial gating**: returns farther than $r_g = 1.2$ m (horizontally) from $\mathbf{c}$ are discarded. This removes on average 50.1% of the remaining returns, almost all of them static clutter. Gating also removes a shortcut: the same furniture appears in training and test recordings, so a model could use the clutter to recognise the scene instead of the motion. MiliPoint is already cropped to the subject and has no Doppler channel, so only median centring is applied. Its axes are remapped to the RadHAR convention (x forward, y lateral, z up) so that both datasets share one coordinate frame.

Temporal segmentation and sampling. A window is split into $G = 12$ equal segments (5 frames on RadHAR at 30 Hz and 4 frames on MiliPoint at 24 Hz). From each segment we sample $P = 32$ returns, with replacement when fewer exist. This gives a fixed tensor $\mathbf{X} \in \mathbb{R}^{G\times P\times C}$ with $C = 6$ features: centred x, y, z; v; ι standardised with training statistics; and the normalised time inside the window. A mask $\mathbf{M} \in \{0, 1\}^{G\times P}$ marks empty segments. Accumulating a few frames per segment counters the extreme sparsity of single frames while keeping temporal order across segments.

Augmentation (training only). Random rotation about the vertical axis ($\pm 15°$), isotropic scaling ($[0.9, 1.1]$), Gaussian

jitter ($\sigma = 1$ cm) and fresh random point sampling at every epoch.

C. STPT-Lite: a factorised spatio-temporal point transformer

Generic point-cloud backbones (PointNet++, DGCNN, Point Transformer) treat the accumulated window as one static set. They spend most of their computation on farthest-point sampling and k-NN neighbourhoods over the whole set, and they encode time only as one more coordinate. Radar HAR has the opposite structure: very few points per instant, but strong regularity over time. STPT-Lite (Fig. 2) therefore factorises attention into a spatial part and a temporal part.

Segment tokenizer. A shared MLP embeds every point, $\mathbf{h} = \phi(\mathbf{x}) \in \mathbb{R}^d$ ($d = 64$). One pre-norm multi-head self-attention block then runs over the P points of each segment, with a learned relative-position bias computed from pairwise 3-D offsets:

$$\mathbf{A}_{ij}^{(h)} = \mathrm{softmax}_j\Big(\frac{\mathbf{q}_i^{(h)} \cdot \mathbf{k}_j^{(h)}}{\sqrt{d/H}} + \beta^{(h)}(\mathbf{x}_i^{xyz} - \mathbf{x}_j^{xyz}) + \log m_j\Big), \tag{1}$$

where β is a two-layer MLP $\mathbb{R}^3 \to \mathbb{R}^H$ and m_j is the mask. The block is followed by a feed-forward layer. The segment token concatenates attentive pooling, $\mathbf{a} = \sum_i \mathrm{softmax}_i(\mathbf{w}^\top \mathbf{h}_i)\mathbf{h}_i$, with max pooling, and projects the result back to d dimensions. Because $P = 32$, full attention inside a segment costs only $G \cdot P^2 = 12{,}288$ pairs, against $(GP)^2 \approx 1.5 \times 10^5$ for joint attention over the window.

Motion descriptor. Radar measures motion directly, so we give it to the model explicitly. For each segment g we compute $\mathbf{u}_g = [\bar{\mathbf{x}}_g,\ \bar{\mathbf{x}}_g - \bar{\mathbf{x}}_{g-1},\ \overline{|v|}_g,\ \log(1 + n_g)] \in \mathbb{R}^8$, i.e. the segment centroid, its displacement since the previous segment, the mean Doppler magnitude and the log point count. An MLP embeds $\mathbf{u}_g$, and the result is added to the segment token.

Temporal encoder. A learnable [CLS] token and positional embeddings are prepended to the G tokens, which pass through $L = 2$ pre-norm Transformer blocks [29] ($H = 4$ heads, FFN width $2d$). Empty segments are masked. The final [CLS] state is layer-normalised and classified. The whole model has 117.5 K parameters and needs 36.6 MFLOPs per 2-s window.

D. Knowledge distillation and quantisation

A wider teacher (STPT-Large: $d = 128$, 4 temporal blocks, 8 heads, 725 K parameters) is trained first. The student minimises

$$\mathcal{L} = (1 - \alpha)\,\mathcal{L}_{\mathrm{CE}}^{\epsilon}(y, p_S) + \alpha\,\tau^2\,\mathrm{KL}\big(p_T^\tau \,\|\, p_S^\tau\big), \tag{2}$$

with label smoothing $\epsilon = 0.1$ [30], $\alpha = 0.7$ and temperature $\tau = 4$ [31]. We also distil into STPT-Tiny ($d = 32$, 30.7 K parameters) to probe the low end of the size range. For deployment, every linear and convolution weight is quantised post-training to INT8, symmetrically and per output channel [32]. We report the accuracy of the quantised model and its storage size.

Noise-robust training. Real deployments see fewer, noisier returns than the recordings. We therefore optionally perturb each training batch: with probability 0.5 each, we apply variable per-segment density (minimum keep fraction 0.3), random point dropout (up to 20%) and false-detection injection (up to 10% of the points), and with probability 0.25 we blank one of the 12 segments. Jitter and severe sparsity are deliberately *not* used in training, so that they test generalisation to unseen corruption types. The test-time perturbation generator uses separate, fixed seeds. The final proposed model, STPT-Lite+KD+robust, combines distillation with this augmentation. It was chosen among the STPT variants by mean validation macro-F1 on both datasets, never by test accuracy.

IV. Experimental Setup

A. Datasets and leakage-free protocols

RadHAR (MMActivity) [1]: TI IWR1443 radar, 30 Hz, five activities (boxing, jumping jacks, jumping, squats, walking), 200 recordings (71.5 min train, 21.4 min test), median 21 returns per frame. We parse the raw ROS dumps and recover the official 71.6/21.4-minute partition. As in [1], samples are 2-s windows (60 frames) with a 10-frame stride (3,565 test windows; the original paper reports 3,538). We keep the *official test recordings* unchanged. The validation set, however, holds out 20% of the *training recordings* per class (29 recordings) instead of 20% of the windows. Neighbouring windows share 50 of 60 frames, so a window-level split would leak near-duplicates into the data used for model selection. A few training recordings are 120–190 s long instead of 20 s, so each validation recording contributes at most 55 evenly spaced windows. This keeps one long recording from dominating model selection (1,595 validation windows). Training windows use a 20-frame stride (4,773 windows), the same for every model.

MiliPoint [6]: TI IWR1843, 24 Hz, 11 subjects, 49 actions. 212,920 frames carry action labels, matching the original paper. The median frame has 6 returns, with xyz only. Labelled frames are grouped into 1,139 contiguous single-action runs, and 2-s windows (48 frames) never cross a run boundary. Our **primary protocol is subject-disjoint**: subjects 8–10 form the test set (3,472 windows), subjects 6–7 the validation set, and the six remaining subjects the training set. For comparison with published numbers we also evaluate (a) the *chronological* 80/10/10 cut used by the MiliPoint code and (b) a *random* 80/10/10 split of overlapping windows, which is leaky by construction.

B. Baselines

All models were re-implemented in pure PyTorch and trained under identical data, augmentation and optimisation settings: the **voxel TD-CNN+BiLSTM** of RadHAR [1] (per-segment $10 \times 32 \times 32$ occupancy grids, 3-D convolutions with 8/16/16 filters, a reduced width for CPU training); **PointNet** [23]; **PointNet+BiLSTM**, a frame-level temporal baseline in the spirit of [10]; **PointNet++** (single-scale grouping) [3]; **DGCNN** [4]; and **Point Transformer** [5]. The static set models receive all $G{\times}P$ points with time as a feature. DGCNN and Point Transformer use a 128-point subset because their

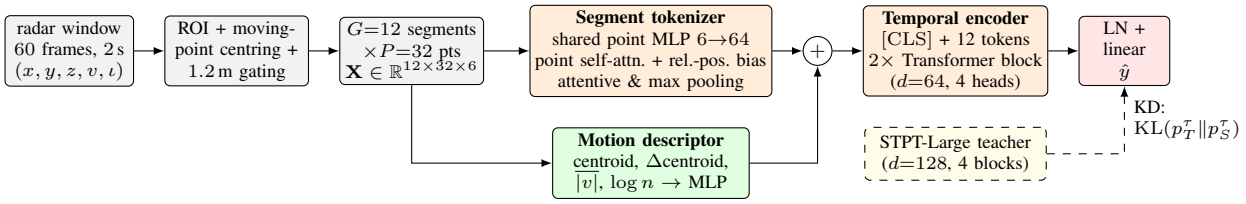


Fig. 2. STPT-Lite. Points are attended to only *within* a short temporal segment (cost $O(GP^2)$), each segment is compressed into one token enriched with an explicit radar motion descriptor, and a small Transformer attends *across* the G segment tokens ($O(G^2)$). No farthest-point sampling or k-NN search is needed. A larger STPT teacher is distilled into the student, whose weights are finally quantised to INT8.

cost grows quadratically with the number of points. We also compare against published results under their own protocols (Table VI).

C. Training and evaluation details

All models use AdamW [33] (learning rate 2×10^{-3}, weight decay 0.05), a one-cycle schedule, batch size 64, label smoothing 0.1 and gradient clipping at 1.0. Training runs for at most 15 epochs with early stopping (patience 5) on *validation macro-F1*. The selected checkpoint is evaluated once on the test set. Every configuration is trained with three seeds (0, 1, 2), and we report mean $\pm$ sample standard deviation. For the reference model we give 95% bootstrap confidence intervals (2,000 resamples of test windows) [34]. Paired comparisons use exact McNemar tests [35] with Holm correction [36] within each seed. Metrics are top-1 accuracy and macro-F1, plus top-3 accuracy for MiliPoint. For efficiency we report parameters, FLOPs per window, FP32/INT8 storage, and median CPU latency at batch size 1 on a single thread (Intel Xeon, PyTorch 2.15). Single-thread CPU latency stands in for an edge processor. All experiments ran on a 2-core CPU without a GPU.

V. Results and Discussion

A. Main comparison

MiliPoint, unseen subjects (Table II). This is the harder and more informative benchmark. The proposed STPT-Lite+KD+robust reaches $49.56 \pm 1.98\%$ top-1 and 78.03% top-3 accuracy on 49 actions (chance $\approx$2%) performed by three people never seen in training or validation (distillation only: $48.55 \pm 4.35\%$). It has the smallest seed spread of all students and comes within one point of its 6× larger teacher ($50.39 \pm 1.39\%$). The strongest re-implemented baseline is PointNet ($32.57 \pm 4.59\%$). Adding a BiLSTM over per-segment PointNet features does *not* close the gap (28.11%), so the gain is not explained by temporal modelling alone. It comes from attending among the points of a segment and from the explicit motion tokens (Section V-B). The difference to every baseline is significant in all three seeds (Holm-corrected McNemar, $p < 10^{-45}$). Our PointNet++ re-implementation performs poorly on MiliPoint (8.07%): with only $\sim$6 returns per frame, its ball queries (radii chosen for RadHAR's denser clouds) yield nearly empty neighbourhoods, and two of three seeds stopped early. We therefore do *not* read this as evidence against PointNet++ in general. The MiliPoint authors report 34.45% for their own implementation under a different, subject-overlapping split [6].

Distillation improves the mean (45.79→48.55%) and, more importantly, the worst case. Without it one seed collapsed to 38.6%, whereas the distilled students stay above 44% in every seed.

RadHAR (Table I). Every model reaches about 95–98%, and the differences between models are mostly within one seed standard deviation. The proposed model obtains $96.83 \pm 0.69\%$ (95% CI of seed 0: [96.52, 97.62]). Plain PointNet is statistically indistinguishable from it in most seeds. RadHAR is saturated under our pre-processing. All models, including the voxel CNN-LSTM of the original paper, exceed the 90.47% reported in [1], which points to the pre-processing (Section V-B) rather than the architecture as the main source of the improvement on this dataset. Our leakage audit also shows that RadHAR's test recordings come from the same sessions and people as its training recordings (Section V-C), so this benchmark cannot measure generalisation to new users.

Efficiency. STPT-Lite has 117.5 K parameters and needs 36.6 MFLOPs per 2-s window (Fig. 3). Its latency is 2.58 ms on one CPU thread, against 7–11 ms for PointNet++, DGCNN and Point Transformer, whose FPS and k-NN steps dominate. With INT8 weights the model occupies 0.14 MB, and accuracy is unchanged (96.82% on RadHAR, 49.59% on MiliPoint). PointNet is faster (0.8 ms) but 16–17 points less accurate on unseen people.

B. Ablation study

Table III removes one component of STPT-Lite (without distillation) at a time and retrains with two seeds on RadHAR. **MiliPoint (unseen people)** is the informative benchmark for the architecture. Compared with the full model trained with the same two seeds (49.4% on average), removing the temporal Transformer (mean over segment tokens instead) costs 11.9 points, and removing the intra-segment attention (pooling only) costs 6.5 points. Both losses hold in each seed separately. Fine-grained actions are therefore distinguished both by how the returns of a short segment relate to each other and by how the segments evolve over the window. The explicit motion descriptor brings no measurable gain (+0.5 points on average, +2.8 and −1.8 in the two seeds): the attention layers apparently recover centroid and displacement information from the point coordinates themselves. We keep the descriptor because it costs only 2.5 K parameters (2%

TABLE I

RADHAR OFFICIAL TEST RECORDINGS (3,565 WINDOWS, 5 CLASSES). MEAN ± STD OVER 3 SEEDS. SIG.: NUMBER OF SEEDS IN WHICH THE HOLM-CORRECTED MCNEMAR TEST AGAINST THE PROPOSED MODEL GIVES $p < 0.05$; ↑ OURS BETTER / ↓ OURS WORSE. LATENCY: MEDIAN, BATCH 1, ONE CPU THREAD.

Model	Params (K)	MFLOPs	Lat. (ms)	Accuracy (%)	Macro-F1 (%)	Sig.
Voxel TD-CNN+BiLSTM [RadHAR]	45.9	180.5	6.60	95.01 ± 0.16	95.03 ± 0.17	2/3↑
PointNet	76.2	31.8	0.74	96.31 ± 0.66	96.36 ± 0.64	1/3↑
PointNet+BiLSTM	109.1	6.6	0.83	96.14 ± 0.23	96.16 ± 0.21	2/3↑
PointNet++ (SSG)	413.4	133.1	8.44	97.34 ± 0.18	97.37 ± 0.19	1/3↓
DGCNN	422.4	137.9	6.95	95.85 ± 0.00	95.89 ± 0.00	1/3↑
Point Transformer	335.2	264.4	7.98	97.01 ± 0.25	97.04 ± 0.24	1/3↑
STPT-Large (teacher)	725.3	139.5	3.59	96.44 ± 1.43	96.48 ± 1.42	1/3↑
STPT-Lite	117.5	36.6	2.14	96.22 ± 1.43	96.27 ± 1.41	2/3↑
STPT-Lite + KD	117.5	36.6	2.08	96.04 ± 0.78	96.08 ± 0.76	1/3↑
STPT-Tiny + KD	30.7	12.1	1.92	96.86 ± 0.17	96.90 ± 0.16	1/3↑
STPT-Lite + KD + robust (ours)	117.5	36.6	2.58	96.83 ± 0.69	96.88 ± 0.68	–
PointNet + noise-robust training	76.2	31.8	0.73	$\mathbf{97.38 \pm 0.11}$	$\mathbf{97.40 \pm 0.11}$	1/3↓

TABLE II

MILIPOINT, SUBJECT-DISJOINT TEST (SUBJECTS 8–10, 3,472 WINDOWS, 49 ACTIONS). SAME CONVENTIONS AS TABLE I.

Model	Params (K)	MFLOPs	Lat. (ms)	Accuracy (%)	Macro-F1 (%)	Top-3	Sig.
PointNet	81.9	31.8	0.74	32.57 ± 4.59	29.72 ± 4.50	59.5	3/3↑
PointNet+BiLSTM	114.8	6.6	0.74	28.11 ± 4.06	28.09 ± 4.66	56.1	3/3↑
PointNet++ (SSG)	424.7	133.1	7.94	8.07 ± 5.47	5.12 ± 5.76	21.4	3/3↑
STPT-Large (teacher)	730.9	139.5	3.65	$\mathbf{50.39 \pm 1.39}$	$\mathbf{50.84 \pm 2.11}$	75.8	1/3↑
STPT-Lite	120.4	36.6	2.60	45.79 ± 6.30	45.95 ± 7.05	73.9	1/3↑
STPT-Lite + KD	120.4	36.6	2.07	48.55 ± 4.35	47.62 ± 6.08	76.4	1/3↓
STPT-Lite + KD + robust (ours)	120.4	36.6	2.00	49.56 ± 1.98	48.80 ± 1.68	78.0	–
PointNet + noise-robust training	81.9	31.8	0.78	18.66 ± 8.83	15.67 ± 8.98	40.9	3/3↑

TABLE III

ABLATION OF STPT-LITE (NO DISTILLATION): TEST ACCURACY (%), MEAN ± STD OVER SEEDS (VARIANTS: 2 ON RADHAR, 2 ON MILIPOINT; FULL MODEL: 3), CHANGE W.R.T. THE FULL MODEL TRAINED WITH THE SAME SEEDS IN BRACKETS; † SINGLE RUN.

Variant	RadHAR	MiliPoint (subj.)
STPT-Lite (full, no KD)	96.22 ± 1.43	45.79 ± 6.30
– intra-segment attention (max+attentive pool only)	96.90 ± 0.18 (+0.93)	42.87 ± 3.03 (-6.52)
– motion descriptor	97.67 ± 0.16 (+1.70)	49.90 ± 2.06 (+0.50)
– temporal Transformer (mean over segments)	96.70 ± 0.61 (+0.73)	37.54 ± 6.62 (-11.85)
– Doppler & intensity (xyz only)	96.42 ± 1.69 (+0.45)	–
– spatial gating	97.62 ± 0.40 (+1.64)	–
– all training augmentation	96.68 ± 0.54 (+0.70)	–
– coordinate jitter augmentation	95.86 ± 1.13 (-0.11)	–

TABLE IV

MILIPOINT (49 ACTIONS) TOP-1 ACCURACY (%) OF THE SAME MODELS UNDER THREE SPLITS. PUBLISHED NUMBERS: 34.45 (POINTNET++, [6]) AND 98.25 (MAK-GCN, RANDOM 80/10/10, [7]).

Model	Subject-disjoint (ours, primary)	Chronological (MiliPoint code)	Random windows (leaky)
STPT-Lite	45.79 ± 6.30	58.38 (n=1)	79.48 (n=1)
PointNet	32.57 ± 4.59	32.42 (n=1)	44.44 (n=1)

of the model), but we do not claim it as a contribution to accuracy. **RadHAR** cannot rank components. Every variant, including removing Doppler/intensity, gating or augmentation, lies within about 1.7 points of the full model, inside its seed spread (± 1.4), because every model already exceeds 95%. The effect of the pre-processing is visible across models instead: with the same gating and centring, all baselines exceed the published 90.47%.

C. Evaluation protocol and leakage

The same STPT-Lite scores 45.79% under our subject-disjoint split, 58.38% under the chronological split of the MiliPoint code (which shares subjects between training and test), and 79.48% under a random split of overlapping windows (Table IV). The automated audit explains this. Under the random split, 72.9% of the test frames also occur inside training windows, and 91.2% of the test windows overlap a training window, by 58.7% of their length on average. The inflation is therefore leakage, not learning. Accuracies of 98% on MiliPoint [7] and ours of about 50% describe different tasks: recognising near-duplicates of seen sequences versus recognising actions of new people. On RadHAR, the same audit found that every official test recording lies within ten minutes of a training recording (median gap 29 s) on the same four days, and the dataset contains no subject identifiers. RadHAR accuracy therefore measures generalisation to new recordings of known people in a known room.

D. Cross-dataset transfer

Table V initialises STPT-Lite from the MiliPoint model (xyz input, harmonised axes) and fine-tunes it on a fraction of the RadHAR training recordings. Pre-training on MiliPoint

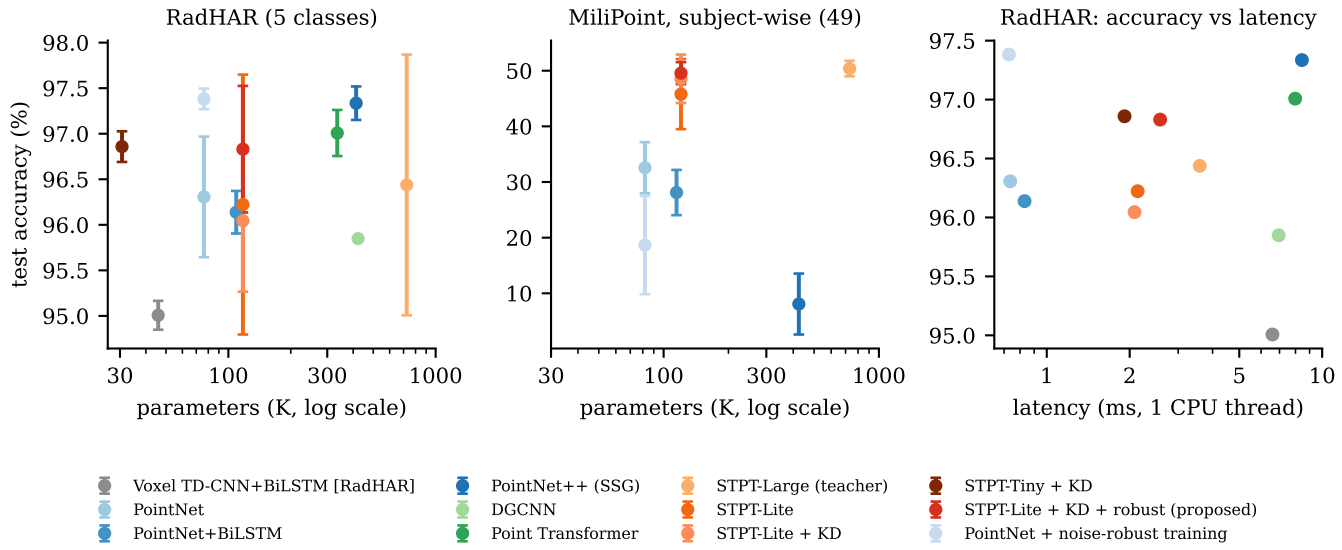


Fig. 3. Accuracy (mean ± std over seeds) versus parameters on both datasets, and RadHAR accuracy versus single-thread CPU latency.

TABLE V
CROSS-DATASET TRANSFER (STPT-LITE, xyz INPUT): TEST ACCURACY (%) WHEN FINE-TUNING ON A FRACTION OF TARGET TRAINING RECORDINGS/RUNS, FROM SCRATCH VS. INITIALISED FROM THE MILIPOINT MODEL (MEAN ± STD OVER 2 SEEDS; N=1 MARKS A SINGLE RUN). SCRATCH AT 100% IS THE xyz-ONLY ABLATION RUN.

Direction	Target data	Scratch	Pre-trained
MiliPoint → RadHAR	10%	61.22 ± 0.58	78.98 ± 4.34
MiliPoint → RadHAR	25%	86.10 ± 1.37	87.45 ± 2.32
MiliPoint → RadHAR	100%	96.42 ± 1.69	95.23 ± 2.02

helps most when little RadHAR data is available. With 10% of the RadHAR training recordings it raises test accuracy from 61.22% to 78.98% (two seeds; +18 points, larger than the seed spread). With 25% the gain shrinks to about one point, inside the seed spread, and with all training data the pre-trained and scratch models are equivalent (95.23% vs. 96.42%). Since the two datasets differ in radar, frame rate, subjects, room and label set, this suggests that the segment tokenizer learns largely sensor-independent motion features from MiliPoint's 49 actions, which are most valuable when target labels are scarce, as in a new home with few annotated recordings. With all data, one pre-trained seed reached its best validation score after the first epoch and was stopped early (93.8%); we report it unchanged.

E. Noise robustness

Fig. 4 reports test accuracy under seven perturbation types, three severities each, applied with fixed seeds so that every model sees the same perturbed windows. **MiliPoint (unseen people).** The proposed model stays the most accurate under every condition except the most extreme sparsity. Noise-robust training improves STPT-Lite+KD under all perturbations, by 0.9–5.4 points (e.g. 20% false detections: 34.6→39.8%; density 0.1: 35.6→39.6%), including jitter, which was not used in training (22.8 vs. 20.5% at $\sigma = 10$ cm). At 12.5% of the returns kept (about one point per frame), STPT-Lite (13.9%) falls to the level of PointNet (14.7%): with almost no points per segment, intra-segment attention has nothing to attend to. **RadHAR.** Dropout, density changes and jitter up to 5 cm cost every point-based model at most 2.5 points. Losing half of the segments costs DGCNN and Point Transformer about five points but the proposed model only 1.2 points (96.8→95.6%), which matches the even temporal attention in Section V-F. Two conditions separate the models. Under extreme sparsity, Point Transformer (92.4%) and PointNet with robust training (91.1%) degrade least, the proposed model reaches 80.0%, and STPT-Lite+KD without robust training drops to 69.7%. False detections are the main weakness of the STPT family on RadHAR: at 20% injected outliers the proposed model keeps 55.1% (39.3% without robust training), whereas PointNet with the same robust training keeps 96.0% and the voxel CNN-LSTM 93.7%. A likely cause is that outliers corrupt the motion descriptor (centroid, mean Doppler) and receive attention weight, while a max-pooled PointNet with robust training learns to ignore isolated points and voxel occupancy dilutes them. Robust training therefore helps, but it does not close this gap. A robust centroid (median) in the motion descriptor and outlier-aware attention are the obvious next steps.

F. Explainability and error analysis

Temporal attention and occlusion. The [CLS] attention of the last block is spread almost evenly over the twelve segments (0.07–0.09 per segment, against 0.083 for uniform attention), with class-specific emphasis: the middle of the window for jumping jacks, early segments for squats, and the first segment

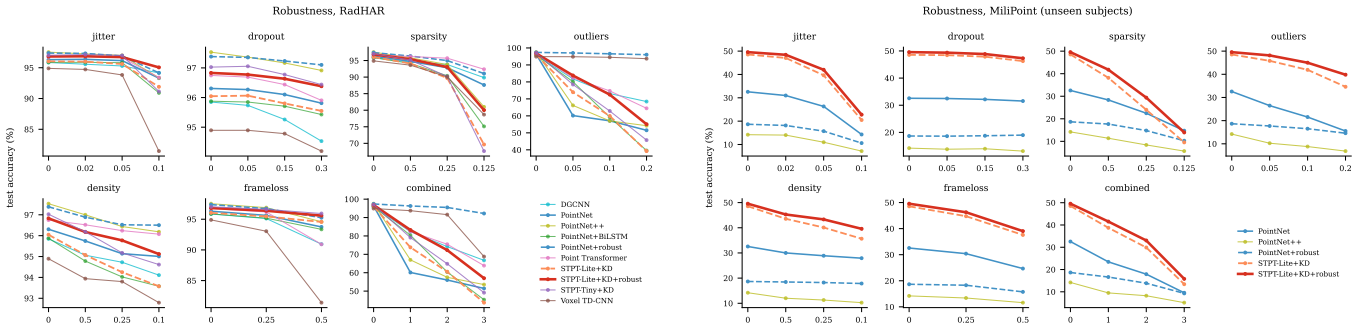


Fig. 4. Test accuracy under controlled perturbations (identical perturbed samples for every model; mean over the available seeds). Left: RadHAR; right: MiliPoint (unseen subjects).

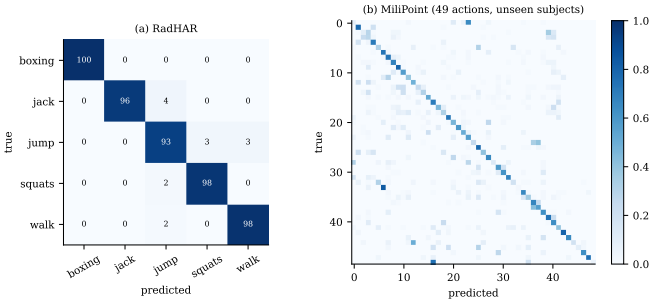


Fig. 5. Row-normalised confusion matrices of the proposed STPT-Lite+KD+robust (seed 0).

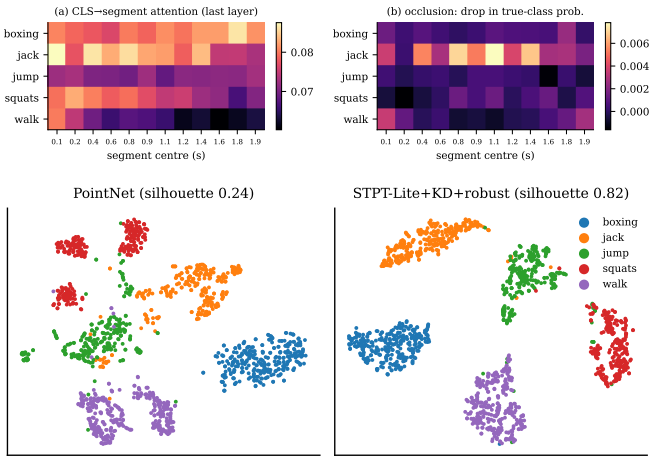


Fig. 6. Top: mean [CLS]→segment attention and segment-occlusion importance per class (RadHAR test). Bottom: t-SNE of test embeddings.

for walking (Fig. 6). Removing a single segment (occlusion) lowers the true-class probability by less than one percentage point on average. The model therefore integrates evidence over the whole window and does not depend on any single 170-ms segment, which is consistent with its tolerance to frame loss. Attention and occlusion importance agree only moderately (Pearson $r = 0.49$ over the class×segment cells), so we treat attention as a coarse indication and not as a faithful explanation. **Point saliency.** Gradient saliency concentrates on the returns of the moving limbs and torso and is low on isolated returns far from the body, i.e. on the residual clutter that survives gating. **Embeddings.** In a t-SNE of RadHAR test embeddings the classes form compact, well-separated clusters (silhouette 0.82 in the model's embedding space versus 0.24 for PointNet). **Errors.** The remaining RadHAR errors are mostly windows predicted as *jump* (precision 91%), which shares the strong vertical Doppler pattern of jumping jacks and squats. On MiliPoint, errors concentrate in groups of kinematically similar actions.

TABLE VI
PUBLISHED RESULTS (AS REPORTED BY THE AUTHORS, UNDER THEIR OWN PROTOCOLS; NOT DIRECTLY COMPARABLE).

Method	Dataset	Acc. (%)	Protocol / note
TD-CNN+BiLSTM [1]	RadHAR	90.47	official test, window-level val.
Lite-PointNet+BiLiLSTM [10]	RadHAR	95.12	79.7 K params
GNN + 3-D joints [9]	RadHAR	96.55	joint pre-training
MAK-GCN [7]	RadHAR	97.54	1.86 M params
OG-PCL [11]	RadHAR	91.75	5-fold CV, 0.83 M params
PointNet++ [6]	MiliPoint	34.45	chronological 80/10/10
MAK-GCN [7]	MiliPoint	98.25	random 80/10/10
STPT-Lite+KD+robust (ours)	RadHAR	96.83	official test, recording-disjoint val.
STPT-Lite+KD+robust (ours)	MiliPoint	49.56	subject-disjoint

Published results. Table VI lists accuracies as reported by other authors. They use different splits, pre-processing and training budgets and are not directly comparable to ours; we include them for context only.

VI. Limitations

Benchmarks. RadHAR has two subjects, no subject identifiers, and test recordings from the same sessions as the training recordings, so it cannot measure generalisation to new users. MiliPoint offers xyz only and three test subjects, and its action names are not published. **Baselines.** All baselines were re-implemented by us under one training budget (at most 15/30 epochs with early stopping, CPU-only). Our PointNet++ did not train well on MiliPoint's six-point frames, and DGCNN and Point Transformer were evaluated on RadHAR only. Published implementations with tuned neighbourhood parameters and longer training may perform better. **Robustness.** Perturbations are applied to recorded point clouds and are not

a physical radar simulation. Under extreme sparsity (12.5% of returns kept), STPT-Lite falls to PointNet level on MiliPoint, and on RadHAR it is far more sensitive to injected false detections than PointNet with the same robust training (55 vs. 96% at 20% outliers). **Deployment.** Latency was measured on one desktop CPU thread, and storage is computed for INT8 weights. On-device deployment was not performed. **Scope.** Single person, fixed 2-s windows, no fall labels, and no clinical validation.

VII. Conclusion and Future Work

We presented STPT-Lite, a factorised spatio-temporal point transformer for privacy-preserving activity recognition from sparse mmWave radar point clouds, together with Doppler-based person centring and clutter gating. On 49 MiliPoint actions performed by unseen people, the distilled, noise-robust model reaches 49.56% top-1 (78.03% top-3), about 17 points above the best re-implemented baseline, at 117.5 K parameters, 36.6 MFLOPs and 2.58 ms per window. On RadHAR all models saturate at about 95–98%, and the pre-processing, not the architecture, explains the gain over the original 90.5%. A dedicated audit shows that random window splits leak most test frames into training, which accounts for much of the spread of published MiliPoint results. Future work: leave-one-subject-out and cross-room evaluation on datasets with subject identifiers and fall events, deployment on the radar's companion processor, and self-supervised pre-training on the large unlabelled portions of both datasets.

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